# LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

## Quick Revision: Acids, Bases and Salts

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### Indicators

| Indicator | In acid | In base |
|---|---|---|
| Blue litmus | turns red | no change |
| Red litmus | no change | turns blue |
| Phenolphthalein | colourless | pink |
| Methyl orange | red | yellow |
| Turmeric | yellow (no change) | reddish-brown |

**Olfactory indicators** (onion, vanilla, clove oil): their smell cannot be detected in a basic solution.

### Key reactions

- Acid + metal → salt + hydrogen: $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$ (gas burns with a pop)
- Base + metal (some): $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2$ (sodium zincate)
- Acid + carbonate/hydrogencarbonate → salt + water + $CO_2$: $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$
- $CO_2$ turns lime water milky: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$
- Neutralisation (acid + base → salt + water): $NaOH + HCl \rightarrow NaCl + H_2O$
- Acid + metal oxide → salt + water: $CuO + 2HCl \rightarrow CuCl_2 + H_2O$ (solution turns blue-green)

### pH and daily life

- pH 0–7 acidic · 7 neutral · 7–14 basic. **Lower pH = stronger acid** (more $H^+$ ions).
- Acids give $H^+$ ($H_3O^+$) ions only in water. Bases soluble in water are called **alkalis** and give $OH^-$ ions.
- Tooth decay starts below pH 5.5. Rain with pH below 5.6 is acid rain.
- Bee sting (methanoic acid): apply baking soda. Indigestion: antacid such as milk of magnesia, $Mg(OH)_2$.

### Important salts

| Salt | Formula | How it's made / key use |
|---|---|---|
| Caustic soda | NaOH | Chlor-alkali process: $2NaCl + 2H_2O \rightarrow 2NaOH + Cl_2 + H_2$ |
| Bleaching powder | $CaOCl_2$ | $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$ · bleaching, disinfecting water |
| Baking soda | $NaHCO_3$ | $NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$ · antacid, baking powder |
| Washing soda | $Na_2CO_3 \cdot 10H_2O$ | Recrystallise sodium carbonate · glass, soap, removes permanent hardness |
| Plaster of Paris | $CaSO_4 \cdot \frac{1}{2}H_2O$ | Heat gypsum at 373 K · plasters for fractures, toys, decoration |

**Water of crystallisation:** copper sulphate crystals $CuSO_4 \cdot 5H_2O$ are blue. On heating they lose water and turn white. Adding water brings back the blue colour.
**Baking powder** = baking soda + a mild edible acid (tartaric acid). The $CO_2$ released makes the cake soft and spongy.

### Common mistakes

- Thinking pH 5 is more acidic than pH 3. It's the other way round.
- Adding water to concentrated acid. Always add **acid to water**, slowly and with stirring.

- Mixing up baking soda ($NaHCO_3$) and washing soda ($Na_2CO_3 \cdot 10H_2O$).

## Quick check

1. Which is more acidic: a solution of pH 3 or one of pH 5?

2. Write the formula of Plaster of Paris and one use.

3. Why does dry HCl gas not change the colour of dry litmus paper?